

AForce OS Spec

Real-Time Human Performance Operating System

May 2026

1. Product Overview

AForce OS is a real-time human performance operating system delivered as a React Native / Expo mobile application backed by an Express 5 + PostgreSQL API server. The product turns hydration into a measurable, scoreable system and surfaces it through a single live “Orb” rather than a dashboard. The platform is designed to scale to 50M+ users.

Core value proposition

- Real-time hydration intelligence with per-event scoring, absorption caps, and physiologically grounded depletion curves.
- A quieter coaching layer: short commands, micro cards, Orb pulse, optional voice — never long monologues.
- Social, environmental, and biometric signals fold into a single score.
- A hidden Recovery Layer plugs underneath every existing surface without redesigning any of them.

Out of scope (by spec, not by oversight)

- No separate Recovery tab, dashboard, or product surface.
- No AI branding, AI chat, or “AI detected” copy anywhere.
- No constant background recalculation — the engine recomputes on discrete events only.

2. Architecture

2.1 Monorepo Layout

- **pnpm workspaces** orchestrate every package.
- Artifacts:
 - artifacts/aforce-os — Expo SDK 54 / React Native mobile app.
 - artifacts/api-server — Express 5 + PostgreSQL + Drizzle ORM.
 - artifacts/aforce-site — marketing site.
 - artifacts/aforce-pitch — investor deck.
- Shared libs in lib/* (composite TypeScript projects).
- OpenAPI contract in @workspace/api-spec; Orval generates Zod schemas + React Query hooks consumed by both server and mobile.

2.2 Mobile (aforce-os)

Layer	Implementation
Framework	Expo SDK 54, Expo Router 6
State	React Context + useReducer (slice-based store)
Data fetching	@tanstack/react-query via generated hooks
Animation	React Native Reanimated 3
i18n	i18next
Auth	@clerk/expo (email/password + Google SSO)
Voice	ElevenLabs primary, Expo Speech fallback
Theme	WHOOOP-Cinematic (pure black #000, lime #B6FF00)

Directory map:

- app/ — root layouts, screens, tabs, gated routes.
- components/ — reusable UI.
- services/ — business logic (engine, hooks, recovery layer).
- store/ — slice-based reducer + selectors.
- utils/ — pure helpers.
- featureFlags/ — flag definitions and demo profiles.
- theme/ — color tokens, typography, spacing, status engine.
- design/ — exported Figma tokens (aforce-tokens.json).
- types/ — global type definitions.
- data/ — mock data, product catalog, templates.

2.3 Server (api-server)

- Express 5 with Zod input/output validation derived from the OpenAPI spec.
- Concurrency-safe writes via SELECT ... FOR UPDATE.
- Structured logging (req.log + singleton logger); no console.log.
- In-memory TTL cache + rate limits for the OpenWeather proxy.
- REST mutations broadcast to connected clients over a shared HTTP / WebSocket server.

2.4 Persistence

- PostgreSQL via Drizzle ORM.
 - Snapshots table carries the optional Recovery Layer columns (recoveryScore, pressureScore, recoveryTrend, recoveryFingerprint, recoveryStory) — all nullable so old rows and flag-OFF writes remain valid.
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3. Identity, Subscription & Commerce

3.1 Authentication

- **Provider:** Clerk. Mobile uses @clerk/expo; server uses @clerk/express middleware.
- **Modes:** custom email/password and Google SSO.
- **Bridging:** Clerk session token is injected into the OpenAPI client so every authenticated mutation carries req.userId.
- **Demo mode:** DEMO_MODE (dev only) bypasses Clerk gates so the welcome flow can be replayed for screenshots.

3.2 Subscriptions & Entitlements

- **Source of truth:** Stripe.
- **Mirror:** stripe-replit-sync writes webhook events to PostgreSQL.
- **Webhook secret:** pulled from the Replit Stripe connector (settings.webhook_secret), not from an env var.
- **Client gating:** useEntitlement.ts hook drives paywalls based on plan tier.
- **Server discipline:** all pricing, shipping, and tax computed server-side; webhook signatures verified.

3.3 Store

- SKU pricing with discounts and bundles.
 - Five consumer subscription tiers with feature gating tied to entitlements.
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4. Core Hydration Engine

- **Per-event scoring** with point values, absorption caps, and release curves.
 - **Depletion math** — a pure, dependency-free helper computes score-points-per-minute decay using physiological standards (sweat rate, climate, activity, body weight).
 - **Water Cycle / “Become AForce”** modals capture water and hydration stick intake with flavor inference.
 - **Social mode** lets alcohol intake feed into score with its own decay curve.
 - **Multi-provider biometrics** (Apple Health, Oura, generic providers) merge via “freshest-wins”.
 - **AForce Protocol screen** derives stage synchronously from current user state (Pause / Hydrate / Lock In / Perform).
 - **Voice Engine** — Cinematic v2 lifecycle state machine, status-color driven, ElevenLabs output, video overlay voices, scripted 60-second investor demo overlay.
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5. Recovery Layer (hidden engine)

5.1 Core rule

Recovery is **not** a separate product. It plugs **underneath** the existing surfaces and feeds them. No new tabs, no redesign, no copy that says “AI”.

5.2 Outputs

The engine produces a snapshot bundle of six values:

Field	Type	Notes
recovery	int 0-100	Clamped
pressure	int 0-100	Clamped
trend	'rising' \ 'stable' \ 'declining'	Energy + decay direction
command	string ≤ 40 chars	Short imperative, no “AI”
fingerprint	8-char lowercase hex	Deterministic on banded inputs
story	string ≤ 120 chars, single line	Free of “AI” / “engine” name-drops

5.3 Inputs

Pulled from existing slices via `recoveryInputsFromState`:

- Hydration score and decay-per-minute (from the scoring engine).
- Water cycles consumed today.
- Urine signal (1-8).
- Heat load and activity level.
- Overnight loss (oz).
- Drink count from `socialMode.drinks`.
- Compliance streak.
- Energy state.

5.4 Surfaces

Surface	Behavior when spec_recovery ON
Orb (home)	Sub-strip below prediction: Recovery N · Pressure N · trend arrow.
Coach	CommandConsole swaps its primary line to recovery.command.
Timeline	Snapshots persist pressure, fingerprint, story.
HydroJournal	Engine reads journal state — Journal UI is unchanged.
Social	“RECOVERY LAYER” section with three tiles (Recovery Cost, Sleep Cost, Recovery Window).
Protocol	Recovery card between stage display and summary rows.
HydroScan	One-line recovery strip beneath ScanResultCard (ellipsized).
Guardian	Phase 2 — Team recovery + recovery story, no personal data.
Clutch	Phase 2 — Pressure, fingerprint, command.
Phantom	<i>(future)</i> Hardware feed for pressure / fingerprint / recovery.
HW	

5.5 Gating contract

- Feature flag: spec_recovery.
 - DEFAULT_FLAGS.spec_recovery = false (production).
 - DEMO_ALL_ON_FLAGS.spec_recovery = true (internal preview).
- useHiddenRecoveryState(...) returns null when the flag is OFF — every surface renders nothing.
- Server-side snapshotSchema keeps the five recovery fields optional, so rows written with the flag OFF leave them NULL.

5.6 Recovery snapshot endpoint

GET /api/aforce/recovery/snapshot

- Auth-gated; resolves userId via requireAuth middleware.
- Returns the latest snapshot row that actually carries Recovery Layer values (filters via recoveryScore IS NOT NULL).
- Response shape:

```
{
  "snapshot": {
    "capturedAt": "ISO-8601",
    "recovery": 0-100,
    "pressure": 0-100,
    "trend": "rising | stable | declining",
    "fingerprint": "8-hex",
    "story": "one-liner"
  }
}
```

or { "snapshot": null } when no qualifying row exists.

5.7 Real-time policy

Event-driven, not streaming. Recompute fires on: water logged, journal entry, sleep update, heat change, HydroScan, workout, social drink, command completed. No interval-based recalculation.

5.8 Release plan

Phase	State
1	Hidden — current state. Engine + 6 surfaces (Orb, Coach, Timeline, HydroJournal, Social, Protocol, HydroScan) wired behind <code>spec_recovery=false</code> .
2	Internal preview. Adds Guardian + Clutch recovery feeds (team recovery + recovery story for Guardian; pressure / fingerprint / command for Clutch).
3	Pressure exposed.
4	Fingerprint exposed.
5	Identity exposed.

The current MVP loop — **Pause → Hydrate → Lock In → Perform** — remains the user-facing system through every phase.

6. UI / UX

6.1 Color & language

- **Canvas:** pure black #000000.
- **Hero accent:** WHOOP lime #B6FF00.
- **Borders:** near-invisible.
- **Status colors:** WHOOP recovery green / yellow / red.
- **Tokens:** exported in Tokens Studio format (`design/aforce-tokens.json`) with a human-readable companion (`design/aforce-design-tokens.md`).

6.2 Design language

- Cinematic dark. Content floats on pure black.
- Data-forward: big numbers, small tracked labels.
- Soft radial glows; never hard box shadows.
- Generous spacing throughout.
- Stylized maps for “Territory”; smooth Reanimated transitions.

7. Server Hardening

- **Routing:** Express 5 with Zod input/output validation derived from the OpenAPI spec.
- **Concurrency:** SELECT ... FOR UPDATE serializes contested writes.
- **Rate limiting & cache:** OpenWeather proxied behind an in-memory TTL cache; public + authenticated endpoints rate-limited.
- **Logging:** structured — `req.log` in handlers, singleton logger elsewhere; `console.log` is forbidden.
- **Realtime:** REST mutations are broadcast over a shared HTTP / WebSocket server.

8. Quality & Testing

- **Vitest** runs unit and integration tests for engine logic, store reducers, server libs, and the recovery store-bridge.
- **TypeScript:** strict; `pnpm run typecheck` is the canonical full check (composite libs build first, then leaf packages).
- **Lint / format:** root-managed Prettier + ESLint.
- **End-to-end:** Playwright-based testing skill runs against the live preview when needed.

The recovery suite alone ships 14 tests across pure derivations (`deriveRecovery`, `derivePressure`, `deriveTrend`, `deriveFingerprint`, `deriveRecoveryCommand`, `deriveRecoveryStory`, `deriveRecoverySnapshot`) and the store-bridge mapper (`recoveryInputsFromState`), including explicit guards

that no output contains the words “AI” or “engine”.

9. Publishing & Distribution

- **Build service:** Expo Application Services (EAS Build), profiles development / preview / production in artifacts/aforce-os/eas.json.
- **Bundle IDs:** iOS com.aforce.os, Android com.aforce.os.
- **App icon / splash / adaptive icon:** WHOOP-cinematic set in assets/images/ (pure black background, lime mark).
- **Build commands** (run inside artifacts/aforce-os/):
 - pnpm eas:build:ios, pnpm eas:build:android, pnpm eas:build:all
 - pnpm eas:build:preview for internal previews.
- **Submit commands:** pnpm eas:submit:ios, pnpm eas:submit:android.
- **One-time iOS submit setup:** run EAS_ASC_APP_ID=... EAS_APPLE_TEAM_ID=... pnpm --filter @workspace/scripts run eas-configure-submit (validated, idempotent, refuses placeholders).
- **One-time Android submit setup:** place Google Play service-account JSON at artifacts/aforce-os/google-service-account.json (never committed).

Store screenshots

App Store and Play Store only accept screenshots from a real device or simulator built from an EAS binary. The Replit web preview cannot produce submission-grade assets — use the iOS Simulator (xcrun simctl io booted screenshot) and Pixel emulator (Android Studio) paths.

10. External Dependencies

Service	Purpose
Stripe	Payments and subscriptions
stripe-replit-sync	Webhook mirroring to Postgres
Clerk	Authentication
Expo SDK 54	React Native runtime
Expo WebBrowser / AuthSession	OAuth, in-app browser
Expo Speech	TTS fallback
@expo-google-fonts/inter	Display font
React Native Reanimated	Declarative animation
React Native Gesture Handler	Gesture recognition
PostgreSQL	Primary database
Drizzle ORM	Schema + query layer
Orval	OpenAPI codegen
Zod	Schema validation
pnpm workspaces	Monorepo management
esbuild	Bundling
OpenWeather	Environmental data
ElevenLabs	Text-to-speech
i18next	Localization

11. Screen Inventory

Every screen shipped in artifacts/aforce-os/app/. Tab routes are the primary navigation surface; everything else is reached via in-app links, deep links, or the activation chain. Items marked **(hidden)** are flagged-off in production.

11.1 Entry & Activation

Route	Purpose
/welcome	Cold-start welcome / re-played in DEMO_MODE; “Begin protocol” CTA.
/(activation)/ready	Pre-flight check before first cycle.
/(activation)/first-command	First scripted command tutorial.
/(activation)/complete	Hand-off to /(tabs); persists hasCompletedOnboarding.
/(auth)/sign-in	Clerk email + Google sign-in.
/(auth)/sign-up	Clerk registration.

11.2 Primary Tabs

Route	Purpose
/ ((tabs)/index)	Home / Orb — single live destination; Recovery Layer sub-strip appears here.
/journal	HydroJournal — longitudinal score, intake, sodium, story line.
/social	Social v2 — circles + RECOVERY LAYER tiles (Recovery / Sleep / Window cost).
/social-legacy	Legacy social mode retained for compatibility.
/protocol	AForce Protocol — Pause / Hydrate / Lock In / Perform stage + recovery card.
/scan	HydroScan — scan result card + one-line recovery strip.
/sleep	Sleep Mode — overnight loss capture (flagged sleep_mode_enabled).
/competition	Sport-mode standings (city / state / team / global).
/profile	Identity, biometrics, providers, DEMO ACCESS toggle, feature-flag matrix.

11.3 Hydration & Health

Route	Purpose
/urine-check	1–8 urine signal capture.
/sweat	Sweat-rate calculator + autopilot toggle.
/heat	Heat-risk surface (climate + activity).
/heat/guardian	Heat alerts feed into Guardian.
/sensors	Sensor / provider import (Apple Health, Oura, ...).
/science	Scientific basis — physiology references behind the engine.

11.4 Coaching, Voice & Demo

Route	Purpose
/notifications	Notification preferences (gated by spec_notifications).
/achievements	Streaks and milestone unlocks.
/share	Generic share entry.
Investor Demo overlay	Scripted 60-second full-screen Voice Engine showcase (not its own route).

11.5 Social Graph

Route	Purpose
/circles	Circles index — private hydration groups.
/circles/[id]	Circle detail with member status board.
/circles/manage	Create / edit circle, invite members.
/circles/shared	Circles the user is a member of but does not own.
/leaderboard	Global / scoped leaderboards.
/territory	Stylized map of the user's "territory" (geo-aggregated activity).

11.6 Commerce

Route	Purpose
/store	Storefront — sticks, bundles, hardware.
/cart	Cart review; server-side pricing / shipping / tax.
/subscription	Subscription tier overview + paywall.
/subscription/manage	Manage active subscription (Stripe customer portal hand-off).

11.7 Premium / Phase 2-3 (flag-gated)

Route	Purpose
/clutch	Clutch Access — team command surface (Phase 2). Recovery feed: pressure / fingerprint / command (Phase 2).
/guardian	Guardian Intelligence — protect-the-roster surface (Phase 2). Recovery feed: team recovery + recovery story (Phase 2).
/phantom	Phantom wearable shell (future hardware feed).
/ring	Ring home — smart-ring companion.
/ring/session	Active ring session.

11.8 Cruise Mode (enterprise add-on)

Route	Purpose
/cruise	Cruise Mode entry surface.
/(hidden)/cruise/journey (hidden)	Journey-Pulse feed (cruise_journey_pulse_enabled).
/(hidden)/cruise/port (hidden)	In-port hydration plan.
/(hidden)/cruise/pre-port (hidden)	Pre-port readiness (cruise_pre_port_enabled).
/(hidden)/cruise/excursion (hidden)	Excursion readiness (cruise_excursion_readiness_enabled).
/(hidden)/cruise/recovery (hidden)	Cruise-scoped recovery view (does not break the no-extra-tab rule).

11.9 Legal

Route	Purpose
/legal/health-disclaimer	Health-claim disclaimer.
/legal/privacy	Privacy policy.
/legal/terms	Terms of service.

11.10 Internal-only surfaces

Reusable screen components in `artifacts/aforce-os/screens/` back the routes

above. Notable internal components without their own route:

- `MySharedStatusScreen` — share-preview composer used by `/share`.
 - `FriendDetailScreen` — friend profile dialog reached from circles.
 - `CruiseModeScreen / (hidden)/cruise/_shared.tsx` — shared cruise chrome (not a route; supporting file).
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12. Glossary

- **Orb** — the single live home surface; the only “dashboard”.
- **Cycle** — one full hydration unit (water + stick + confirmation).
- **Pressure** — accumulated physiological load (heat, activity, decay, alcohol), released by cycles.
- **Fingerprint** — deterministic 8-hex hash of banded recovery inputs; stable identity across micro-fluctuations.
- **Recovery story** — one-line, single-sentence narrative attached to a snapshot. Never references AI or the engine.
- **Spec flag** — a `spec_*` flag gating a new architecture surface; OFF in production by default.